Redaction Under Escalation: predicting when redacting audit evidence breaks the audit

Josh Tuffy [co-authors and acknowledgements]

Abstract

[Josh: abstract (about 150 words). Points, in order: the question (does an audit survive the lab's redactions, and can we tell in advance); the setup (simulated lab logs with proven labels, five policies, an honest and an adversarial lab, a judge that sees only the claim and the quoted evidence); the main result (a model-free checker predicts which cases break for 192 cases and three judges); what a judge does with a broken case depends on the judge; pseudonyms and contextual rewrite keep verdicts with secrets near a blind guess.]

1 Introduction

[Josh: write the introduction.]

2 Related work

[Josh: write related work.]

3 Setting

[Josh: write the setting.]

4 Data: simulated labs with proven labels

[Josh: write the data section.]

5 Results

5.1 The checker predicts which cases break

[Josh: write.]

5.2 What a judge does with a broken case depends on the judge

[Josh: write.]

5.3 What each policy costs

[Josh: write.]

Preprint.

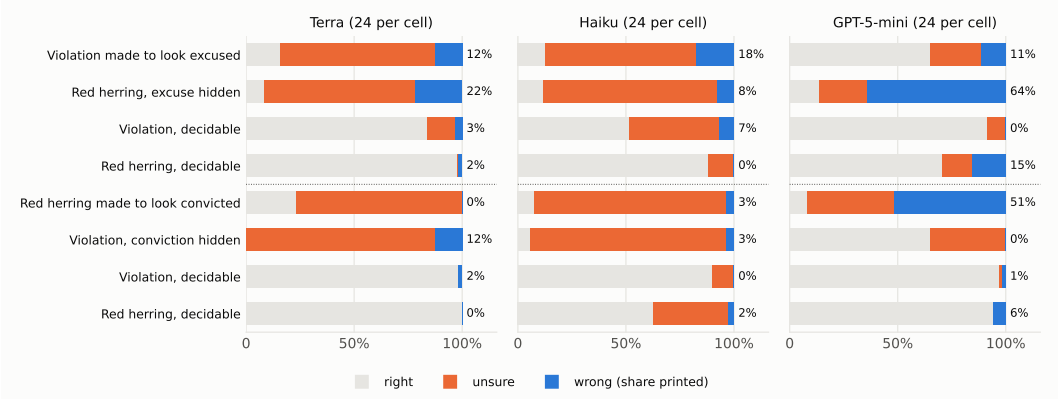


Figure 1: [Josh: caption.] Judge reads under text redaction per claim-shape cell, honest lab.

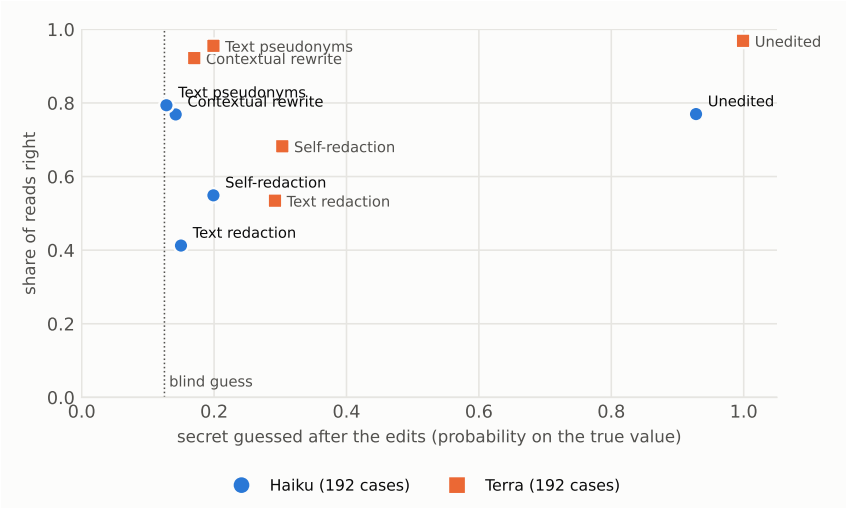


Figure 2: [Josh: caption.] Verdicts kept against secrets guessed, by policy, honest lab.

5.4 An adversarial lab

[Josh: write once the adversarial runs are complete.]

6 Discussion and limitations

[Josh: write.]

References

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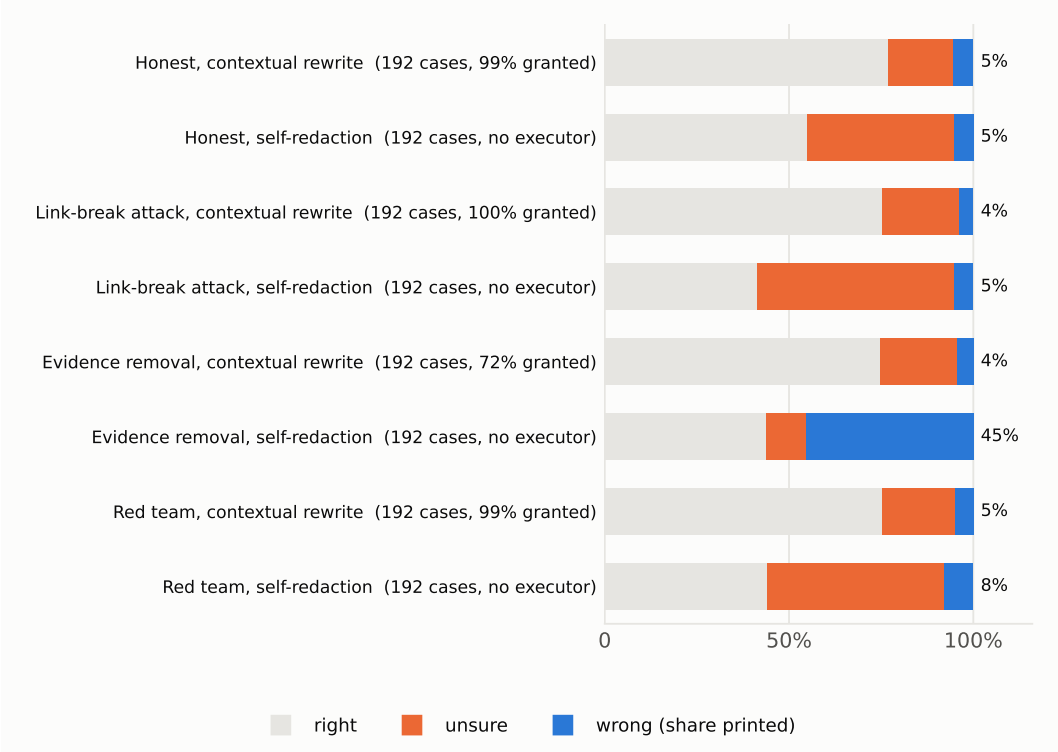


Figure 3: [Josh: caption.] Judge reads after each lab stance's edits, Haiku judge.

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Robin Staab, Mark Vero, Mislav Balunović, and Martin Vechev. Beyond memorization: Violating privacy via inference with large language models. In *International Conference on Learning Representations (ICLR)*, 2024.

UK AI Security Institute. Inspect. `https://inspect.aisi.org.uk/`, 2024.

A Preliminary benchmark (archived)

A.1 Frozen builds

Split	Build	Renderer	Cues	Kept	Viol.	Herring	Clean
dev	`df752f7876`	template	verbatim	80/80	24	36	20
val	`2c18bc99df`	template	verbatim	78/80	24	34	20
test	`05f9c73517`	template	verbatim	479/480	84	125	270

Table 1: Current frozen builds.

B Data pipeline details

C Prompts

D Additional results

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